Final Review Script (about 5 minutes)

Slide 1 - Ritheek

Good morning. This is the final review of our project, Intelligent Traffic Signal Control Using Fuzzy Logic and Optimization. Since Review 1 we have moved from a pitch to a working prototype with measured results. Today we show how it works, how we tested it, and what the results show.

Slide 2 - Srilakshmi

Most traffic signals run on a fixed timer, so every direction gets the same green time whatever is waiting. One road can be packed while the other is empty, and drivers pay for it in delay. Our controller sets green time from live queue and waiting-time data instead.

Slide 3 - Srilakshmi

Since Review 1 we built the full fuzzy controller, a queue simulator, genetic-algorithm tuning and an interactive 3D replay. Most recently we built version 2, with a stronger controller and a much more rigorous evaluation. All four objectives are complete: controller, tuning, benchmark and prototype.

Slide 4 - Deekshan

The controller takes three inputs: the queue on its own road, its waiting time, and the queue on the opposing road. Fuzzification turns each into low, medium or high, and 27 IF-THEN rules combine them. Centroid defuzzification gives one output, a green time between 10 and 60 seconds. The prototype shows which rules fired, so every decision is explainable.

Slide 5 - Deekshan

Version 1 only saw its own queue and wait, so it was blind to the opposing road. Version 2 adds the opposing queue, grows from 9 to 27 rules, and tunes all membership breakpoints instead of just three centres. Testing also grew from 3 hand-picked scenarios to 120 generated ones.

Slide 6 - Priyakaran

There is no real-world dataset, so the data is simulated traffic and we say so. We generated 120 scenarios with rush-hour ramps, surges and lulls, split into 40 to train, 20 to validate and 60 to test, scored once. We compared a

genetic algorithm, particle swarm and simulated annealing with the same budget. Fixed seeds make every result reproducible.

Slide 7 - Priyakaran

Here is how we trained it. Fuzzy logic decides the green time, and the genetic algorithm tunes it. Each candidate is a list of membership numbers that becomes a fuzzy controller. We run it in the simulator on the 40 training scenarios, and its average delay is its score. The GA keeps the best, crosses and mutates them, within 2000 evaluations. We pick the winner on 20 validation scenarios and score it once on 60 unseen ones.

Slide 8 - Priyakaran

This is a working prototype, not only a report. A command-line tool returns the green time and the rules that fired, and a local web app runs the real simulator and opens the 3D replay. There you can switch between the fixed timer and our controller at the same second. 23 tests pass.

Slide 9 - Madhan

These results are on the 60 test scenarios nothing was tuned on. Against a fixed 30-second timer, version 1 cuts delay by 34.9 percent and version 2 by 58.5 percent. Version 2 beats version 1 by 42.4 percent, winning 59 of 60 scenarios.

Slide 10 - Madhan

We also tested a simple actuated rule, green equals 3 seconds per queued vehicle. It cuts delay by 57.8 percent, so version 2 beats it by only 4.3 percent and loses 14 of 60 scenarios. The choice of optimizer made no real difference. Limits: simulated data and one two-phase intersection.

Slide 11 - Ritheek

To conclude: adaptive control cuts delay by about 58 percent against a fixed timer, with rules we can explain. Version 2 improves version 1 by 42 percent. Limits: simulated data and one two-phase intersection. Thank you, we are happy to take questions.